

Leader–Follower Affine Formation Control of Second-Order Nonlinear Uncertain Multi-Agent Systems

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Abstract—This brief proposes two leader-follower affine formation control algorithms for a class of second-order uncertain multi-agent systems. The model of each agent is described by the second-order dynamics with nonlinear uncertainties. Due to the high robustness, flexibility and maneuverability, the affine formation approaches based on the stress matrix are adopted to achieve a target formation, in which the formation control laws are designed for followers to keep a desired formation configuration with respect to leaders. First, considering that the velocities of leaders are constant, we design an estimator to compensate for the uncertainties and propose a PI-like affine formation control law to guarantee that the followers can achieve the target formation. Then, for the case that the velocities of leaders are time-varying, we propose another control algorithm with variable gains for followers under which a time-varying formation configuration can be maintained. Finally, the corresponding simulations of a realistic example are carried out to validate the effectiveness of the proposed affine formation control algorithms.

Index Terms—Uncertain multi-agent systems, affine formation control, variable gain control.

I. INTRODUCTION

THE FORMATION control problem of multi-agent systems (MASs) has attracted much attention due to its wide applications in many fields such as unmanned vehicles, etc. The aim of formation control is to design control laws to drive agents to their desired states such that the target formation can be achieved. Formation approaches can be divided into leaderless [1], single-leader [2], multi-leader cases. In the more challenging multi-leader case [3], [4], the aim is to drive the followers into some desired spaces spanned by leaders, which has more flexibility and maneuverability because the whole formation can be easily controlled by controlling a small part of agents, namely, leaders. Multi-leader

case is more preferable in engineering practices since it has more robustness against the failure of a subsystem of MASs and flexibility of adding new members into MASs. There are mainly two types of multi-leader formation approaches, namely, formation-containment control and affine formation control, which are based on the Laplacian matrix [5] and the stress matrix [6], respectively. Since the Laplacian matrix is a special case of the stress matrix, the followers should be in the convex hull spanned by leaders in the formation-containment case while the followers can be outside the convex hull in the case of affine formations. Moreover, the stress-matrix-based affine formation is invariant to several affine transformations such as translation, rotation, scaling, shearing [6]. Due to this high maneuverability of the affine formation approach, the formation can be reshaped according to the affine transformation indicated by leaders. Motivated by these advantages, this brief employs the stress-matrix-based leader-follower affine formation approach to achieve the target formation.

Up to now, there have been many studies on the formation control of various uncertain models, while the current results on the stress matrix-based leader-follower affine formation control rarely take the uncertainty into account. In fact, there are many kinds of uncertainties and disturbances in the practical systems [7]. The consensus problem [8] and formation-containment control problem with consideration of uncertainties have been extensively researched, while the affine formation control has not. Many physical systems are subject to the uncertain disturbances which are difficult to measure and are usually nonlinear. Thus, our task in this brief is to design control laws for followers governed by second-order dynamics with nonlinear uncertainties. Furthermore, some proposed control laws have some limitations on the states of leaders such as static or constant moving velocity. However, in contrast to the static formation cases, the time-varying formation problems have more applications in practice. The time-varying formations of unmanned aerial vehicle (UAV) swarm systems [9] and autonomous underwater vehicle systems [10] have broad potential applications in civilian and military areas, such as drag reduction, contour mapping, target search and localization, target enclosing, and so on. Moreover, the time-varying velocity of leaders has less restriction which makes the formation configuration more flexible. Thus, we consider affine formation control with leaders of time-varying velocities.

In conclusion, using the stress-matrix-based leader-follower affine formation approach, we will propose the new variable gain control law for second-order dynamics with uncertainties, which can achieve the time-varying affine formation. To authors' best knowledge, the time-varying affine formation

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control has not been fully investigated, especially for agents with nonlinear dynamics. Thus, the main contributions of this brief can be summarized as follows.

(i) The paper [11] has studied the affine time-varying formation control of multi-agent systems with triple-integrator dynamics. Different from [11], we consider the more general second-order dynamics with nonlinear uncertainties in practice.

(ii) The affine formation control problems with uncertainty are mainly limited to the case of constant velocity, while we take both the cases of constant velocity and time-varying velocity into consideration, which have more applications in practice.

II. NOTATION AND PRELIMINARIES

A. Notation and Basic Graph Theory

Denote $\otimes$ as the Kronecker product and define $\circ$ as the Hadamard product. Denote $\lambda_{\max}(\cdot)$ and $\lambda_{\min}(\cdot)$ as the largest and smallest eigenvalue of a matrix, respectively. Let 0_d be the zero matrix in $\mathbb{R}^{d \times d}$. Define $\mathbf{1}_d$ as the $d \times 1$ column vector of all ones. Denote $\text{sgn}(\cdot)$ as the sign function. Define $\text{abs}(\cdot)$ as the absolute function which gets the matrix with absolute value of each element. We say $f \in \mathbb{L}_h$ when $(\int_0^\infty |f(\tau)|^h d\tau)^{1/h}$ exists for $h \in [1, \infty)$ and $f \in \mathbb{L}_\infty$ when $\sup_{t \geq 0} |f(t)|$ exists.

In this brief, consider multi-agent systems consisting of n leaders and m followers. Assume that all followers are governed by the following nonlinear dynamics

$$\begin{aligned} \dot{p}_i &= v_i \\ \dot{v}_i &= u_i + \Phi_i \theta_i, i = 1, \dots, m \end{aligned} \quad (1)$$

where $p_i \in \mathbb{R}^d$ denotes agent i 's position, $v_i \in \mathbb{R}^d$ denotes agent i 's velocity, $u_i \in \mathbb{R}^d$ is the input of agent i , $\Phi_i(p_i, v_i) \in \mathbb{R}^{d \times q}$ is a matrix whose elements are bounded and continuous functions of p_i and v_i . $\theta_i \in \mathbb{R}^q$ is the unknown and constant parameter vector of agent i , e.g., the mass or the fluid density in the environment. We denote the group of leaders as $\mathcal{V}_l = \{m+1, \dots, m+n\}$. Define the position of followers as $p_f = [p_1^T, \dots, p_m^T]^T$ and the position of leaders as $p_l = [p_{m+1}^T, \dots, p_{m+n}^T]^T$.

An undirected graph $\mathcal{G} = (\mathcal{V}, \varepsilon)$ is used to describe the interaction among agents, which consists of a vertex set $\mathcal{V} = \{1, \dots, m+n\}$ and edge set $\varepsilon \subseteq \mathcal{V} \times \mathcal{V}$. The set of neighbors of vertex i is denoted by $\mathcal{N}_i = \{j \in \mathcal{V} : (i, j) \in \varepsilon\}$. A formation in $\mathbb{R}^d$ is denoted by $(\mathcal{G}, x)$ which is the graph $\mathcal{G}$ equipped with a configuration x in $\mathbb{R}^d$ with the vertex i corresponding to point p_i . Denote the constant configuration $r = [r_1^T, \dots, r_{m+n}^T]^T$ as a nominal configuration. Define the nominal formation as $(\mathcal{G}, r)$. Then, the target formation with maneuvers can be described by $p^*(t) = [I_{m+n} \otimes E(t)]r + \mathbf{1}_{m+n} \otimes b(t)$, where $p^*(t)$ is the target position, $b(t)$ represents the formation's translational maneuver and $E(t)$ can be used to realize the whole formation's rotational, scaling, and shearing maneuvers with respect to the nominal configuration r . Thus, the desired position of agent i in the target formation is $p_i^*(t) = E(t)r_i + b(t)$.

B. Stress Matrix

For a formation $(\mathcal{G}, p)$, a stress represents a set of scalars $\{\omega_{ij}\}_{(i,j) \in \varepsilon}$ with $\omega_{ij} = \omega_{ji} \in \mathbb{R}$, which is assigned to all the edges. A stress is called an equilibrium if it satisfies

$\sum_{j=1}^{m+n} \omega_{ij}(p_i - p_j) = 0$, which can be described in the form of $(\Omega \otimes I_d)p = 0$, where $\Omega \in \mathbb{R}^{(m+n) \times (m+n)}$ is the stress matrix satisfying

$$\Omega_{ij} = \begin{cases} 0, & \text{if } i \neq j, (i, j) \notin \varepsilon \\ -\omega_{ij}, & \text{if } i \neq j, (i, j) \in \varepsilon \\ \sum_{k \in \mathcal{N}_i} \omega_{ik}, & \text{if } i = j \end{cases} \quad (2)$$

Partition Ω according to the partition of leaders and followers as $\Omega = \begin{bmatrix} \Omega_{ff} & \Omega_{fl} \\ \Omega_{lf} & \Omega_{ll} \end{bmatrix}$, where $\Omega_{ff} \in \mathbb{R}^{m \times m}$, $\Omega_{fl} \in \mathbb{R}^{m \times n}$, $\Omega_{lf} \in \mathbb{R}^{n \times m}$ and $\Omega_{ll} \in \mathbb{R}^{n \times n}$. To simplify the calculation, define $\Omega_{ff} \otimes I_d$ as $\tilde{\Omega}_{ff}$.

Definition 1 (Affine Span [6]): Given a set of points $\{p_i\}_{i=1}^{m+n}$ in $\mathbb{R}^d$, the affine span S of these points is

$$S = \left\{ \sum_{i=1}^{m+n} a_i p_i : a_i \in \mathbb{R} \text{ for all } i \text{ and } \sum_{i=1}^{m+n} a_i = 1 \right\}.$$

The affine image of the nominal configuration is defined as

$$\mathcal{A}(r) = \left\{ p \in \mathbb{R}^{d(m+n)} : p = (I_{m+n} \otimes E)r + \mathbf{1}_{m+n} \otimes b \right\}.$$

Definition 2 (Affine Localizability [6]): The nominal formation $(\mathcal{G}, r)$ is affinely localizable by leaders if for any $p = [p_f^T, p_l^T]^T \in \mathcal{A}(r)$, p_f can be uniquely determined by p_l .

Assumption 1: We assume that

- (1) the nominal configuration r affinely span $\mathbb{R}^d$.
- (2) the nominal formation $(\mathcal{G}, r)$ has a positive semi-definite stress matrix Ω satisfying $\text{rank}(\Omega) = m+n-1$.
- (3) the nominal formation $(\mathcal{G}, r)$ is affinely localizable by the leaders.

Lemma 1 [6, Th. 2]: Under Assumption 1, $\tilde{\Omega}_{ff}$ is positive definite, and for any $p = [p_f^T, p_l^T]^T \in \mathcal{A}(r)$, p_f can be uniquely calculated as

$$p_f = -(\tilde{\Omega}_{ff}^{-1} \Omega_{fl} \otimes I_d) p_l. \quad (3)$$

III. AFFINE FORMATION CONTROL FOR LEADERS WITH CONSTANT VELOCITY

In this section, we assume that the velocity of leaders is constant, i.e., $\dot{v}_i = 0, i \in \mathcal{V}_l$ and our task is to design a control law for followers to achieve the target formation, i.e., $\lim_{t \rightarrow \infty} (p_f(t) - p_f^*(t)) = \lim_{t \rightarrow \infty} (p_f(t) + (\tilde{\Omega}_{ff}^{-1} \Omega_{fl} \otimes I_d) p_l(t)) = 0$, where $p_f^* \in \mathbb{R}^{dm}$ is the desired position vector for all followers. For simplicity, we define the formation error $\xi_i(t) = p_i(t) - p_i^*(t)$. Let $\xi_f = [\xi_1^T, \dots, \xi_m^T]^T = p_f + (\tilde{\Omega}_{ff}^{-1} \Omega_{fl} \otimes I_d) p_l$.

First, we design the following estimators to eliminate the effect of the unknown parameter θ_i

$$\dot{\hat{p}}_i = \hat{v}_i - (\hat{p}_i - p_i) \quad (4)$$

$$\dot{\hat{v}}_i = u_i + \Phi_i \hat{\theta}_i - (\hat{p}_i - p_i) - (\hat{v}_i - v_i) \quad (5)$$

$$\dot{\hat{\theta}}_i = -k_{oi} \Phi_i^T (\hat{v}_i - v_i) \quad (6)$$

where $\hat{p}_i \in \mathbb{R}^d$, $\hat{v}_i \in \mathbb{R}^d$, $\hat{\theta}_i \in \mathbb{R}^q$ are the estimation of p_i , v_i , θ_i respectively and $k_{oi} \in \mathbb{R}$ is the positive constant gain.

First, we define the estimation errors of position, velocity and unknown parameter respectively as $\tilde{p}_i = \hat{p}_i - p_i$, $\tilde{v}_i = \hat{v}_i - v_i$ and $\tilde{\theta}_i = \hat{\theta}_i - \theta_i$. Using the fact (1), we can obtain the dynamics of the estimation errors

$$\dot{\tilde{p}}_i = \tilde{v}_i - \tilde{p}_i \quad (7)$$

$$\dot{\tilde{v}}_i = \Phi_i \tilde{\theta}_i - \tilde{p}_i - \tilde{v}_i \quad (8)$$

$$\dot{\tilde{\theta}}_i = -k_{oi} \Phi_i^T \tilde{v}_i \quad (9)$$

Then, we consider the Lyapunov function as

$$V_{ei} = \frac{1}{2} \tilde{p}_i^T \tilde{p}_i + \frac{1}{2} \tilde{v}_i^T \tilde{v}_i + \frac{1}{2k_{oi}} \tilde{\theta}_i^T \tilde{\theta}_i \quad (10)$$

Taking the time-derivative of the Lyapunov function, we can have

$$\dot{V}_{ei} = \tilde{p}_i^T \dot{\tilde{p}}_i + \tilde{v}_i^T \dot{\tilde{v}}_i + \frac{1}{k_{oi}} \tilde{\theta}_i^T \dot{\tilde{\theta}}_i = -\tilde{p}_i^T \tilde{p}_i - \tilde{v}_i^T \tilde{v}_i \leq 0 \quad (11)$$

which implies that $\dot{V}_{ei}$ is negative semi-definite and $\tilde{p}_i, \tilde{v}_i, \tilde{\theta}_i$ are uniformly bounded. Differentiating (11) yields

$$\dot{V}_{ei} = -\tilde{p}_i^T \dot{\tilde{p}}_i - \tilde{v}_i^T \dot{\tilde{v}}_i = -\tilde{p}_i^T (\tilde{v}_i - \tilde{p}_i) - \tilde{v}_i^T (\Phi_i \tilde{\theta}_i - \tilde{p}_i - \tilde{v}_i) \quad (12)$$

Therefore, $\dot{V}_{ei}$ is bounded and consequently, $\dot{V}_{ei}$ is uniformly continuous in time. Since V_{ei} is lower bounded, $\dot{V}_{ei}$ is negative semi-definite and uniformly continuous, using the Lyapunov-like lemma [12, Lemma 4.3], we can have that $\dot{V}_{ei}(t) \rightarrow 0$ as $t \rightarrow \infty$, which indicates that $\lim_{t \rightarrow \infty} \tilde{p}_i(t) = 0$, $\lim_{t \rightarrow \infty} \tilde{v}_i(t) = 0$, $i = 1, \dots, m$.

We define the position tracking error and the velocity tracking error as $e_{pi} = \hat{p}_i - p_i^*$ and $e_{vi} = \hat{v}_i - v_i^*$, where p_i^* and v_i^* represent the target position and velocity of agent i , respectively. According to Lemma 1, we can get the tracking errors in a compact form

$$e_p = \hat{p}_f - p_f^* = \hat{p}_f + (\Omega_{ff}^{-1} \Omega_{fl} \otimes I_d) p_l \quad (13)$$

$$e_v = \hat{v}_f - v_f^* = \hat{v}_f + (\Omega_{ff}^{-1} \Omega_{fl} \otimes I_d) v_l \quad (14)$$

where $e_p = [e_{p1}^T, \dots, e_{pm}^T]^T$, $e_v = [e_{v1}^T, \dots, e_{vm}^T]^T$, $p_l = [p_{m+1}^T, \dots, p_{m+n}^T]^T$, $v_l = [v_{m+1}^T, \dots, v_{m+n}^T]^T$.

Then, we design the following control law

$$\begin{aligned} u_i(t) = & -\Phi_i(p_i, v_i) \hat{\theta}_i(t) + \tilde{p}_i(t) + \tilde{v}_i(t) \\ & -k \left(\sum_{j=1}^{m+n} \omega_{ij} (\hat{p}_i(t) - \hat{p}_j(t)) + \sum_{j=1}^{m+n} \omega_{ij} (\hat{v}_i(t) - \hat{v}_j(t)) \right. \\ & + k_p \int_0^t \sum_{j=1}^{m+n} \omega_{ij} (\hat{p}_i(\tau) - \hat{p}_j(\tau)) d\tau \\ & \left. + k_v \int_0^t \sum_{j=1}^{m+n} \omega_{ij} (\hat{v}_i(\tau) - \hat{v}_j(\tau)) d\tau \right) \end{aligned} \quad (15)$$

where $k, k_p, k_v \in \mathbb{R}$ are positive constant gains and $\hat{p}_j = p_j, \hat{v}_j = v_j, \forall j \in \mathcal{V}_l$. Writing (15) into a compact form yields

$$\begin{aligned} u_f(t) = & \tilde{p}_f(t) + \tilde{v}_f(t) - k \left(\bar{\Omega}_{ff} e_p(t) + k_p \int_0^t \bar{\Omega}_{ff} e_p(\tau) d\tau \right. \\ & \left. + \bar{\Omega}_{ff} e_v(t) + k_v \int_0^t \bar{\Omega}_{ff} e_v(\tau) d\tau \right) - \Phi_f(p_f, v_f) \hat{\theta}_f(t) \end{aligned} \quad (16)$$

where $\hat{\theta}_f = [\hat{\theta}_1^T, \dots, \hat{\theta}_m^T]^T$, $\tilde{p}_f = [\tilde{p}_1^T, \dots, \tilde{p}_m^T]^T$, $\tilde{v}_f = [\tilde{v}_1^T, \dots, \tilde{v}_m^T]^T$, $\Phi_f = \text{diag}\{\Phi_1, \dots, \Phi_m\}$.

Theorem 1: Consider the system (1) with the estimators (4)–(6), and suppose that Assumption 1 holds. If

$$\lambda_{\min}(\bar{\Omega}_{ff}) > \frac{k_p}{k(k_v + 1)} \quad (17)$$

is satisfied, then the control law (15) can guarantee that the followers will converge to their desired positions, i.e., $\lim_{t \rightarrow \infty} \xi_f(t) = 0$.

Proof: Using the control law (15), one has the dynamics of tracking errors

$$\dot{e}_p = e_v - \tilde{p}_f \quad (18)$$

$$\begin{aligned} \dot{e}_v = & u_f + \Phi_f \hat{\theta}_f - \tilde{p}_f - \tilde{v}_f + \left(\Omega_{ff}^{-1} \Omega_{fl} \otimes I_d \right) \dot{v}_l \\ = & -k \bar{\Omega}_{ff} \left(e_p + k_p \int_0^t e_p d\tau + e_v + k_v \int_0^t e_v d\tau \right) \\ & + (\Omega_{ff}^{-1} \Omega_{fl} \otimes I_d) \dot{v}_l \end{aligned} \quad (19)$$

$$\begin{aligned} \ddot{e}_v = & -k \bar{\Omega}_{ff} (k_p e_p + \dot{e}_p + k_v e_v + \dot{e}_v) \\ = & -k \bar{\Omega}_{ff} ((k_v + 1) e_v + k_p e_p + \dot{e}_v - \tilde{p}_f) \end{aligned} \quad (20)$$

Therefore, we can have the following augmented tracking error dynamics:

$$\begin{aligned} \frac{d}{dt} \begin{bmatrix} e_p \\ e_v \\ \dot{e}_v \end{bmatrix} = & \begin{bmatrix} 0_{dm} & I_{dm} & 0_{dm} \\ 0_{dm} & 0_{dm} & I_{dm} \\ -kk_p \bar{\Omega}_{ff} & -k(k_v + 1) \bar{\Omega}_{ff} & -k \bar{\Omega}_{ff} \end{bmatrix} \begin{bmatrix} e_p \\ e_v \\ \dot{e}_v \end{bmatrix} - \begin{bmatrix} I_{dm} \\ 0_{dm} \\ -k \bar{\Omega}_{ff} \end{bmatrix} \tilde{p}_f \\ = & A_e \begin{bmatrix} e_p \\ e_v \\ \dot{e}_v \end{bmatrix} - \begin{bmatrix} I_{dm} \\ 0_{dm} \\ -k \bar{\Omega}_{ff} \end{bmatrix} \tilde{p}_f \end{aligned} \quad (21)$$

Defining the eigenvalue of A_e as λ , one has the characteristic polynomial of matrix A_e

$$\begin{aligned} |\lambda I_{3dm} - A_e| &= \left| \lambda^3 I_{dm} + \lambda^2 k \bar{\Omega}_{ff} + k(k_v + 1) \lambda \bar{\Omega}_{ff} + kk_p \bar{\Omega}_{ff} \right| \\ &= \left| \left(\frac{\lambda^3 I_{dm}}{\lambda^2 k + k(k_v + 1) \lambda + kk_p} + \bar{\Omega}_{ff} \right) \right| \\ &= \left| \left(\lambda^2 k + k(k_v + 1) \lambda + kk_p \right) \right| \end{aligned} \quad (22)$$

Letting $|\lambda I_{3dm} - A_e| = 0$, one has that $\lambda^2 k + k(k_v + 1) \lambda + k \times k_p = 0$ and $\lambda^3 + k \sigma \lambda^2 + k \sigma (k_v + 1) \lambda + kk_p \sigma = 0$ where σ is the eigenvalue of $\bar{\Omega}_{ff}$. According to the Routh-Hurwitz criterion, if $k \sigma (k_v + 1) - k_p > 0$, the eigenvalues of A_e all have the negative real part. Because k, k_p and k_v are positive constant gains, it follows that when (17) holds, the matrix A_e is Hurwitz. Considering that $\lim_{t \rightarrow \infty} \tilde{p}_f(t) = 0$, we can have that the tracking error (13), (14) can converge to zero according to input-to-state theorem [13], i.e., $\lim_{t \rightarrow \infty} (\hat{p}_f(t) + (\Omega_{ff}^{-1} \Omega_{fl} \otimes I_d) p_l(t)) = 0$, $\lim_{t \rightarrow \infty} (\hat{v}_f(t) + (\Omega_{ff}^{-1} \Omega_{fl} \otimes I_d) v_l(t)) = 0$. Using $\lim_{t \rightarrow \infty} \tilde{p}_f(t) = 0$, we can have that $\lim_{t \rightarrow \infty} (p_f(t) + (\Omega_{ff}^{-1} \Omega_{fl} \otimes I_d) p_l(t)) = 0$. This completes the proof. ■

Remark 1: In this brief, the nonlinear parametric uncertainties $\Phi_i \theta_i$ exist directly in the second-order dynamics (1) which is more challenging than the parametric uncertainty in first-order closed-loop dynamics [14]. To handle this challenge, different from [14], we first design position and velocity estimators in (4)–(6) to compensate for these uncertainties.

IV. AFFINE FORMATION CONTROL FOR LEADERS WITH TIME-VARYING VELOCITY

In this section, we assume the velocity of leaders is continuous time-varying and the acceleration $\dot{v}_i(t)$, $i \in \mathcal{V}_l$ is bounded.

We propose the following algorithm for followers

$$u_i = -s_i - \hat{\gamma}_i \circ \text{sgn}(s_i) - \Phi_i \hat{\theta}_i \quad (23)$$

$$\dot{\hat{\gamma}}_i = c_1 \text{abs}(s_i) \quad (24)$$

$$\dot{\hat{\theta}}_i = c_2 \Phi_i^T s_i \quad (25)$$

where $s_i = \sum_{j=1}^{m+n} (p_i - p_j) + a \sum_{j=1}^{m+n} (v_i - v_j)$, $\hat{\theta}_i \in \mathbb{R}^q$, $\hat{\gamma}_i \in \mathbb{R}^d$ and $a, c_1, c_2 \in \mathbb{R}$ are all positive constant gains.

Theorem 2: Considering the system (1), suppose that Assumption 1 holds and the acceleration of leaders is bounded and the velocity of leaders is continuous. If

$$\lambda_{\min}(\bar{\Omega}_{ff}) > a^{-3} \quad (26)$$

is satisfied, then the control law (23)-(25) can guarantee that the followers will converge to their desired positions, i.e., $\lim_{t \rightarrow \infty} \xi_f(t) = 0$.

Proof: First, define $e_i = \sum_{j=1}^{m+n} (p_i - p_j)$. Therefore, we can have $s_i = e_i + a\dot{e}_i$. Consider the Lyapunov function as

$$V = \frac{1}{2} \left\{ \frac{a(\hat{\gamma}_f - \gamma \mathbf{1}_{dm})^T (\hat{\gamma}_f - \gamma \mathbf{1}_{dm})}{c_1} + s_f^T \bar{\Omega}_{ff}^{-1} s_f + e_f^T \bar{\Omega}_{ff}^{-1} e_f + \frac{a(\theta_f - \hat{\theta}_f)^T (\theta_f - \hat{\theta}_f)}{c_2} \right\} \quad (27)$$

where $s_f = [s_1^T, \dots, s_m^T]^T$, $\hat{\gamma}_f = [\hat{\gamma}_1^T, \dots, \hat{\gamma}_m^T]^T$, $\gamma \in \mathbb{R}$ is the positive constant to be decided later, $\hat{\theta}_f = [\hat{\theta}_1^T, \dots, \hat{\theta}_m^T]^T$, $\theta_f = [\theta_1^T, \dots, \theta_m^T]^T$, $e_f = [e_1^T, \dots, e_m^T]^T$.

Taking the time-derivative of s_f , we can have

$$\dot{s}_f = \dot{e}_f + a(\bar{\Omega}_{ff}(-s_f - \hat{\gamma}_f \circ \text{sgn}(s_f) + \Phi_f(\theta_f - \hat{\theta}_f)) + (\Omega_{ff} \otimes I_d)\dot{v}_l) \quad (28)$$

Combining (27) with (24), (25) and (28), one gets

$$\begin{aligned} \dot{V} &= a(\hat{\gamma}_f - \gamma \mathbf{1}_{dm})^T \text{abs}(s_f) + (e_f + a\dot{e}_f)^T \bar{\Omega}_{ff}^{-1} \dot{e}_f \\ &\quad - a s_f^T s_f - a s_f^T \text{sgn}(s_f) \circ \hat{\gamma}_f + a s_f^T (\Omega_{ff}^{-1} \Omega_{ff} \otimes I_d) \dot{v}_l \\ &\quad + a s_f^T \Phi_f(\theta_f - \hat{\theta}_f) + e_f^T \bar{\Omega}_{ff}^{-1} \dot{e}_f - a(\theta_f - \hat{\theta}_f)^T \Phi_f^T s_f \\ &= -a\gamma \mathbf{1}_{dm}^T \text{abs}(s_f) + a s_f^T (\Omega_{ff}^{-1} \Omega_{ff} \otimes I_d) \dot{v}_l - a s_f^T s_f \\ &\quad + (e_f + a\dot{e}_f)^T \bar{\Omega}_{ff}^{-1} \left(\dot{e}_f + \frac{1}{a} e_f - \frac{1}{a} e_f \right) + e_f^T \bar{\Omega}_{ff}^{-1} \dot{e}_f \\ &= -a\gamma \mathbf{1}_{dm}^T \text{abs}(s_f) + a s_f^T \ddot{p}_f^* - a s_f^T s_f - \frac{1}{a} e_f^T \bar{\Omega}_{ff}^{-1} e_f \\ &\quad + \frac{1}{a^2} (e_f + a\dot{e}_f)^T \bar{\Omega}_{ff}^{-1} (e_f + a\dot{e}_f) \\ &\leq a(\|\ddot{p}_f^*\|_1 - \gamma) \|s_f\|_1 - \frac{1}{a} e_f^T \bar{\Omega}_{ff}^{-1} e_f \\ &\quad - s_f^T (aI_{dm} - \frac{1}{a^2} \bar{\Omega}_{ff}^{-1}) s_f \end{aligned} \quad (29)$$

where $\ddot{p}_f^* \in \mathbb{R}^{dm}$ represents the target accelerations of all followers. Because $\bar{\Omega}_{ff}$ is positive definite, it follows that

$$aI_{dm} - \frac{1}{a^2} \bar{\Omega}_{ff}^{-1} = aP \begin{bmatrix} 1 - \frac{1}{a^3 \lambda_1} & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 1 - \frac{1}{a^3 \lambda_{dm}} \end{bmatrix} P^T \quad (30)$$

where $P \in \mathbb{R}^{dm \times dm}$ is the unit orthogonal matrix of matrix $\bar{\Omega}_{ff}^{-1}$ and $\lambda_i > 0$, $i = 1, \dots, dm$ is the eigenvalue of $\bar{\Omega}_{ff}$.

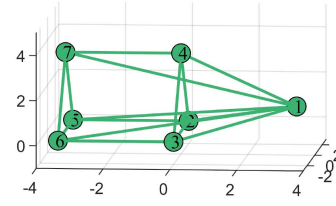


Fig. 1. Nominal formation communication graph.

Because (26) holds, we can have that the matrix $(aI_{dm} - \frac{1}{a^2} \bar{\Omega}_{ff}^{-1})$ is positive definite. Considering that $\dot{v}_l$ is bounded, we can have $\ddot{p}_f^*$ is also bounded. Therefore, if γ is sufficiently large such that $\|\ddot{p}_f^*\|_1 \leq \gamma$ holds, one gets that $\dot{V}$ is negative semi-definite, which implies that s_f and e_f are both bounded. Since $V \geq 0$ and $\dot{V} \leq 0$, there always exists $V(\infty) \in [0, V(0)]$ such that $\lim_{t \rightarrow \infty} V(t) = V(\infty)$. By integrating (29) on the both sides, we can have that

$$\int_0^\infty a(\gamma - \|\ddot{p}_f^*\|_1) \|s_f\|_1 dt + \int_0^\infty s_f^T \left(aI_{dm} - \frac{1}{a^2} \bar{\Omega}_{ff}^{-1} \right) s_f dt + \int_0^\infty \frac{1}{a} e_f^T \bar{\Omega}_{ff}^{-1} e_f dt \leq V(0) - V(\infty) \quad (31)$$

which implies that $s_f \in \mathbb{L}_2 \cap \mathbb{L}_\infty$ and $e_f \in \mathbb{L}_2 \cap \mathbb{L}_\infty$. According to the Barabara's lemma [12, Lemma 4.2], one has that $\lim_{t \rightarrow \infty} e_f(t) = 0$, which implies that $\lim_{t \rightarrow \infty} \xi_f(t) = 0$. This completes the proof. ■

Remark 2: The constant γ in (27) is only used for the sake of proof which has no physical meaning and will not be used in the control implementation. Moreover, the variable gain $\hat{\gamma}_i$ has an effect on the system convergence and robustness.

Remark 3: Since the control law (23) has the sign function which induces chattering, we can reduce chattering by choosing c_1 small enough. Moreover, one can also use of a tunable low-pass filter (LPF) to avoid the chattering [15].

V. SIMULATION

In this section, we consider the real situation where UAVs receive the disturbance of air resistance [16]. When the UAV is flying at high speed, the air resistance is proportional to the square of the velocity, which can be described by Kv_i^2 . Therefore, we select Φ_i as the diagonal matrix with diagonal elements given as $-v_{i(h)}^2$, $h = 1, 2, 3$. Then, θ_i includes the unknown information of the air density, drag coefficient, etc. Consider the nominal formation as shown in Fig. 1. The leaders are marked as the shaded nodes 1-4, all the other nodes are followers. The corresponding positive semi-definite stress matrix is given as following

$$\Omega = \begin{bmatrix} 0.375 & -0.25 & -0.25 & -0.25 & 0.125 & 0.125 & 0.125 \\ -0.25 & 0.333 & 0.083 & 0.083 & -0.25 & 0 & 0 \\ -0.25 & 0.083 & 0.333 & 0.083 & 0 & -0.25 & 0 \\ -0.25 & 0.083 & 0.083 & 0.333 & 0 & 0 & -0.25 \\ 0.125 & -0.25 & 0 & 0 & 0.208 & -0.042 & -0.042 \\ 0.125 & 0 & -0.25 & 0 & -0.042 & 0.208 & -0.042 \\ 0.125 & 0 & 0 & -0.25 & -0.042 & -0.042 & 0.208 \end{bmatrix}$$

A. Simulation of the Affine Formation Control Law for Leaders With Constant Velocity

Assume the velocities of leaders are constant. Note that $\lambda_{\min}(\bar{\Omega}_{ff}) = 0.125$. According to Theorem 1, the gains of control law (15) are selected as $k = 35$, $k_p = 15$, $k_v = 45$. The simulation results are given by Fig. 2, Fig. 3(a) and Fig. 4(a),

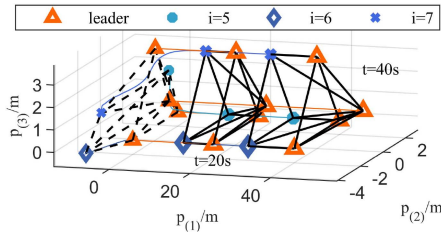


Fig. 2. The trajectories of agents under the control law (15).

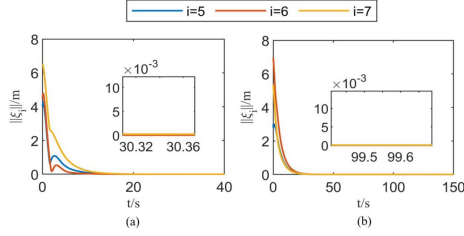


Fig. 3. The evolution of the norm of the formation error ξ_i for each follower over time by the control law (15) (a) and (23) (b) respectively.

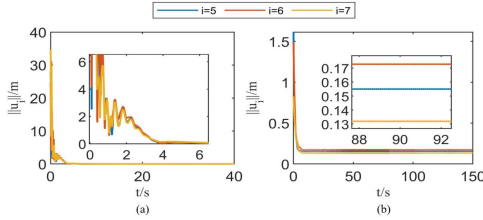


Fig. 4. The evolution of the norm of the control law (15) (a) and (23) (b) for each follower over time respectively.

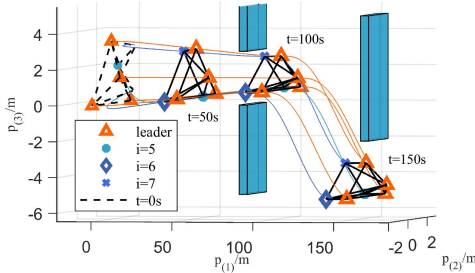


Fig. 5. The trajectories of agents using the control law (23).

which show that the error ξ_i converges to zero asymptotically and the followers can move to their target positions quickly.

B. Simulation of the Affine Formation Control Law for Leaders With Time-Varying Velocity

Now, we consider the case of time-varying leaders. According to Theorem 2, the gains of the control law (23) are given as $a = 5$, $c_1 = 0.1$, $c_2 = 10$.

The simulation results are shown in Fig. 5, Fig. 3(b) and Fig. 4(b) which give the trajectories of agents with a time-varying formation configuration, the tracking errors ξ_i of followers and the control law u_i , respectively. We can see that the chattering is within an acceptable range. This simulation indicates that the stress-matrix-based affine formation control method has high maneuverability to change the formation by

only controlling followers to follow leaders which can perform translation, rotation and scaling. Note that the state in these formation control problems is general which can also be regarded as the voltage in real circuits [17].

VI. CONCLUSION

In this brief, the affine formation control laws for the second-order multi-agent systems with parameter uncertainties have been designed. First, we have designed the control algorithm for leaders with constant velocity to achieve the target formation. Then, considering the velocities of leaders are time-varying, we proposed the control law with variable gains. Finally, the simulation results have illustrated the effectiveness of the presented approaches. Future work will concentrate on the affine formation control under directed graph.

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